

Regression

Irene Siragusa, PhD

Outline

-  Regression modelling
 -  Linear regression
 -  Ridge regression
 -  Lasso regression
 -  Principal Component Regression
 -  Generalized Linear Models
 -  Non-linear regression
-  Regression evaluation

Regression Modelling with Numeric Classes

- What if class variables are numeric?
- The objective is to minimize the squared error of prediction of the numeric class variable
 - Linear regression
 - Principal component regression
 - Generalized linear models
 - Nonlinear and Polynomial Regression
 - Regression trees

Linear regression

- Given a data set D of n vectors in $\mathbb{R}^d$ $D = \begin{bmatrix} \bar{X}_1 \\ \vdots \\ \bar{X}_n \end{bmatrix}$, $(n \times d)$ and given n matches $X_i \rightarrow y_i, i = 1, \dots, n$
- Determine a vector of coefficients $W = \{w_1, \dots, w_d\}$ which allows the approximation of the matches $X_i \rightarrow y_i, i = 1, \dots, n$, as:

$$y_i \approx \bar{W} \cdot \bar{X}_i \quad \forall i \in \{1 \dots n\}$$

Linear regression

- The objective function, defined as the Sum of Squared Errors (SSE) is minimized:

$$O = \sum_{i=1}^n (\bar{W} \cdot \bar{X}_i - y_i)^2 = \|D\bar{W}^T - \bar{y}\|^2$$

- Thus it is referred to as Least Squares Regression
 - The Fisher linear discriminant is a special case of this if:
 - D is a mean zero
 - $y_i = -1/p_0$ for negative class
 - $y_i = 1/p_1$ for positive class

Regularization in regression

- A regularization in the objective function may be added to prevent overfitting
- Smaller coefficients are desirable to reduce overfitting
- A regularization parameter λ is introduced
- Ridge regression $\rightarrow$ L_2 regularization
- Lasso regression $\rightarrow$ L_1 regularization

Ridge regression

- Ridge regression

$$O = \|D\bar{W}^T - \bar{y}\|^2 + \lambda \|\bar{W}\|^2$$

$$\bar{W}^T = (D^T D + \lambda I)^{-1} D^T \bar{y}$$

- L₂ regularization
- $D^T D$ is positive semidefinite and therefore always invertible
- It reduces the variance of the estimate
- $\lambda > 0$ should be selected via an adaptively cross-validation

Lasso regression

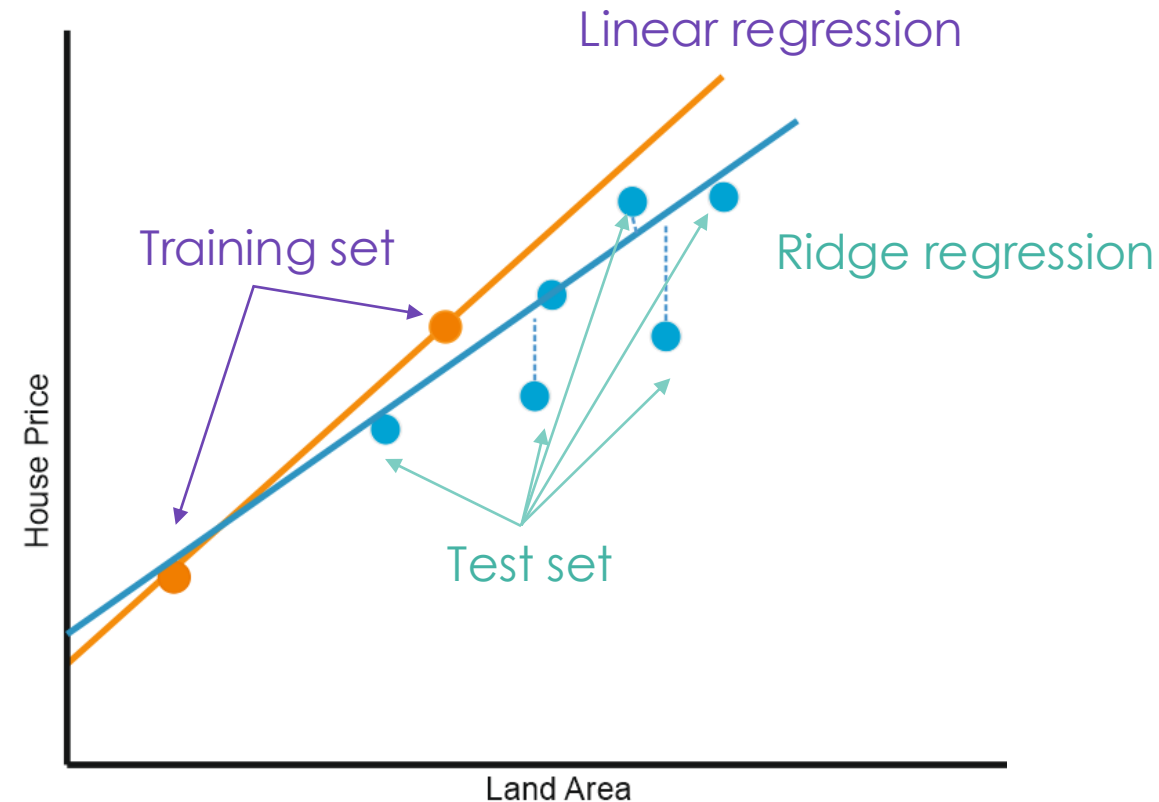
- Lasso regression

$$O = \|D\bar{W}^T - \bar{y}\|^2 + \lambda \|\bar{W}\|$$

- L_1 regularization
- It tends to select sparse solutions
- A vector of the scattered coefficients is preferred
- Suitable in the case of the presence of many irrelevant features for the given problem, corresponding coefficients will be reset
 - Highly interpretable feature selection process

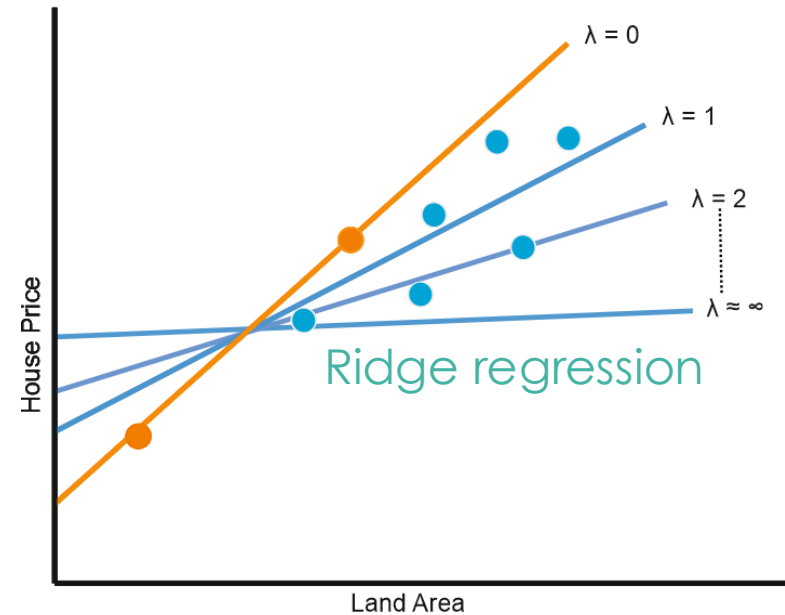
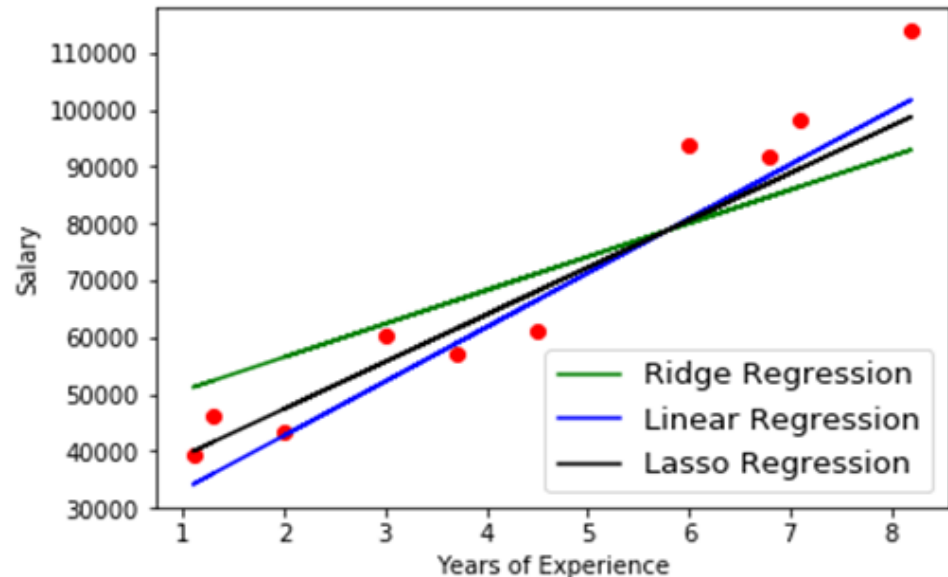
Regularization in regression

- Regularization effect



Regularization in regression

- Ridge vs Lasso
 - Ridge regression only asymptotically eliminates the contribution of a feature ($\lambda \rightarrow \infty$) while the Lasso regression can eliminate the feature exactly (zero slope for λ very large, but not ∞)



Principal Component Regression

- Used to reduce the dimensionality of the data set compared to uninteresting or noisy features
- PCA is applied to D thus obtaining a new data matrix $R = \begin{bmatrix} \bar{Z}_1 \\ \vdots \\ \bar{Z}_n \end{bmatrix}$, $(n \times k)$ projecting D over k principal directions
- Thus obtaining a (computational) reduced algorithm as in linear regression

$$y_i \approx \bar{W} \cdot \bar{Z}_i \quad \forall i \in \{1 \dots n\}, \quad \bar{W} = (R^T R)^{-1} R^T \bar{y}$$

Generalized Linear Models

- In generalized regression linear models, a mean function is estimated $f(\bar{W} \cdot \bar{X}_i)$, $i = 1, \dots, n$ as from the probability distribution function of numerical labels y_i
 - Each response variable is modeled as an outcome of a probability distribution function
- Characteristics of linear regression are kept, but are extended to diverse data types
- The objective is then:

$$y_i \sim \text{Probability distribution with mean } f(\bar{W} \cdot \bar{X}_i) \quad \forall i \in \{1 \dots n\}$$

Generalized Linear Models

- The choice of the mean function, or its inverse $f^{-1}(\cdot)$ called link function, underlies a particular probability distribution and therefore a certain type of data

	Link function	Mean function	Distribution assumption
Least squares regression	Identity	$\bar{W} \cdot \bar{X}$	Normal
	Inverse	$-1/(\bar{W} \cdot \bar{X})$	Exponential, Gamma
	Log	$\exp(\bar{W} \cdot \bar{X})$	Poisson
Logistic regression	Logit	$1/[1 + \exp(-\bar{W} \cdot \bar{X})]$	Bernoulli, Categorical
	Probit	$\Phi(\bar{W} \cdot \bar{X})$	Bernoulli, Categorical

Generalized Linear Models

- A maximum likelihood approach is used to determine the optimal parameter set $\bar{W}$
 - The product of the probabilities (or probability densities) of the response variable outcomes is maximized.
- After estimating the parameters in $\bar{W}$, the expected response value of a test instance $\bar{T}$ is estimated as $f(\bar{W} \cdot \bar{T})$.

Non Linear Regression

- Extension of the regression to a non linear function of the sample:

$$y = \sum_{i=1}^m w_i h_i(\bar{X})$$

where $h_i(\cdot)$ represents a non linear transformation function from d -dimensional input feature space to 1-d space

- Capture complex functional relationships between data set points

Non Linear Regression

- Non Linear Regression

- In polynomial regression, the higher powers of each dimension up to order r are used as a new set of derived features.
- This approach expands the number of dimensions, but it allows greater expressiveness in terms of nonlinear relationships.
- The main disadvantage is the dimensionality expansion that may result in overfitting, which can be prevented with regularization techniques.

- Kernel ridge regression

- Kernel trick is used to model relations between data and numerical labels

Regression trees

- Non linear relations among data are modelled as a set of local linear approximation in the leaf nodes of the tree.
- The same split criteria and pruning techniques, as for decision trees, are used.
- For numerical data, the value of SSE over all resulting nodes obtained after splitting, are used.
- A regression is required at each node
 - Alternatively, the variance of data after each split can be calculated and regression is applied only to leaf nodes *after* the tree has been built

Regression evaluation

- Via R^2 statistics o R^d

$$R^2 = 1 - \frac{SSE}{SST}$$

$$R^2 = 1 - \frac{(n - d)}{(n - 1)} \frac{SSE}{SST}$$

$$SSE = \sum_{i=1}^n (y_i - g(\overline{X_i}))^2$$

$$SST = \sum_{i=1}^n \left(y_i - \sum_{j=1}^n \frac{y_j}{n} \right)^2$$

Sum of Squared Errors (SSE)

Sum of Squares Total (SST)